



## 1. Advanced Log Analysis

### Introduction:

Advanced log analysis is the process of collecting, examining, correlating, and interpreting logs from multiple systems to identify security threats, suspicious behavior, system failures, and policy violations.

**Security Operations Center (SOC) analysts use advanced log analysis to:**

- Detect cyberattacks
- Investigate incidents
- Reduce false positives
- Understand attacker behavior

**Logs come from many sources:**

- Firewalls
- Endpoints
- Servers
- Applications
- IDS/IPS

### 1.1 Log Correlation

Log correlation is the process of connecting events from different log sources to identify meaningful attack patterns.

#### Example:

- Windows logs show multiple failed logins (Event ID 4625)
- Firewall logs show suspicious outbound traffic
- Together, this may indicate account compromise and data exfiltration.

#### Common Log Sources

| Source             | Purpose                     |
|--------------------|-----------------------------|
| Firewall Logs      | Network activity            |
| Windows Event Logs | Login/authentication events |
| EDR Logs           | Endpoint behavior           |



| Source     | Purpose        |
|------------|----------------|
| DNS Logs   | Domain lookups |
| Proxy Logs | Web traffic    |

## Correlation Techniques

### 1.Rule-Based Correlation

Uses predefined rules.

#### Example:

- IF 5 failed logins occur within 2 minutes
- AND successful login follows
- THEN trigger brute-force alert

### 2.Time-Based Correlation

Matches events occurring within a specific timeframe.

### 3.Behavioral Correlation

Analyzes user or system behavior patterns.

#### Example:

- User normally logs in from India
- Suddenly logs in from another country
- SIEM flags anomaly

### 1.2 Anomaly Detection

Anomaly detection identifies unusual behavior that differs from normal system activity.

## Types of Anomalies

### 1.User Behavior Anomalies

Examples:

- Login at unusual hours
- Excessive failed logins
- Unusual file access



## 2. Network Anomalies

Examples:

- High outbound traffic
- Connections to rare domains
- Unexpected protocol usage

## 3. System Anomalies

Examples:

- Sudden CPU spikes
- Unauthorized software installation
- Privilege escalation

## Detection Methods

### Statistical Analysis

Uses baseline metrics.

Example:

- Average login attempts = 3/day
- Current activity = 50 attempts/hour
- Alert generated

### Rule-Based Detection

Uses predefined thresholds.

Example:

- Alert if data transfer exceeds 5GB

### Machine Learning Detection

Uses AI models to detect abnormal behavior patterns.

## 1.3 Log Enrichment

Log enrichment means adding additional contextual information to raw logs.

### Types of Log Enrichment

#### 1. Geolocation Enrichment

Adds country/city details for IP addresses.



## 2. Threat Intelligence Enrichment

Matches IPs, domains, and hashes against threat feeds.

## 3. Asset Enrichment

Adds:

- Device owner
- Department
- Asset criticality

## 4. User Context Enrichment

Adds:

- User role
- Privilege level
- Business unit

## Benefits of Log Enrichment

| Benefit                   | Description                             |
|---------------------------|-----------------------------------------|
| Faster investigations     | Analysts get immediate context          |
| Better prioritization     | Critical assets receive higher priority |
| Reduced false positives   | Better event understanding              |
| Improved threat detection | IOC matching improves visibility        |

## 2. Threat Intelligence Integration

### Introduction

Threat intelligence provides information about current and emerging cyber threats. SOC teams use this intelligence to improve detection, response, and proactive threat hunting.

### 2.1 Types of Threat Intelligence

#### 1. Strategic Intelligence

High-level information for executives.

Examples:



- Threat trends
- Industry-targeted attacks

## **2.Tactical Intelligence**

Information about attacker techniques.

Examples:

- Phishing methods
- Malware delivery techniques

## **3.Operational Intelligence**

Details about ongoing attacks.

Examples:

- Attack campaigns
- Threat actor infrastructure

## **4.Technical Intelligence**

Machine-readable indicators.

Examples:

- Malicious IPs
- File hashes
- Domains

## **Tactics, Techniques, and Procedures (TTPs)**

TTPs describe attacker behavior patterns.

### **Tactics**

High-level attack goals.

### **Techniques**

Methods used to achieve goals.

### **Procedures**

Specific implementation steps.

## **STIX and TAXII**



## **STIX**

Structured Threat Information eXpression.

Used for:

- Sharing threat intelligence
- Standardizing IOC data

## **TAXII**

Trusted Automated eXchange of Indicator Information.

Used for:

- Automated intelligence sharing

## **2.2 Threat Intelligence Integration in SIEM**

### **Workflow**

#### **Step 1: Collect Threat Feeds**

Import IOC feeds into SIEM.

#### **Step 2: Enrich Logs**

Compare logs against threat feeds.

#### **Step 3: Generate Alerts**

Trigger alerts when matches occur.

#### **Step 4: Investigate and Respond**

Analysts validate and respond to threats.

## **3. Incident Escalation Workflows**

### **Introduction**

Incident escalation workflows ensure security incidents are handled efficiently by the correct SOC teams.

Proper escalation:

- Reduces response time
- Improves coordination
- Minimizes business impact

### **3.1 SOC Tier Structure**

#### **Tier 1 – Alert Triage**



## **Responsibilities**

- Monitor alerts
- Validate false positives
- Initial analysis
- Create incident tickets

## **Skills Required**

- SIEM monitoring
- IOC identification
- Basic investigation

## **Tier 2 – Incident Investigation**

### **Responsibilities**

- Deep investigation
- Malware analysis
- Threat containment
- Root cause analysis

### **Skills Required**

- Log analysis
- Threat hunting
- Endpoint analysis

## **Tier 3 – Advanced Analysis**

### **Responsibilities**

- Advanced forensics
- Threat intelligence
- Detection engineering
- Incident recovery support

### **Skills Required**

- Reverse engineering
- Advanced threat analysis
- SIEM tuning



## Escalation Criteria

| Criteria        | Description                     |
|-----------------|---------------------------------|
| Severity        | Critical or High alerts         |
| Scope           | Multiple systems affected       |
| Complexity      | Requires advanced investigation |
| Business Impact | Sensitive systems involved      |

## Escalation Workflow Process

### Step 1: Detection

SIEM generates alert.

### Step 2: Triage

Tier 1 validates alert.

### Step 3: Escalation

Incident assigned to Tier 2 or Tier 3.

### Step 4: Investigation

Detailed analysis performed.

### Step 5: Containment

Threat isolated.

### Step 6: Recovery

Systems restored.

### Step 7: Lessons Learned

Post-incident review conducted.

## 3.2 Communication Protocols

### SITREP (Situation Report)

Components

- Incident summary
- Affected systems
- Indicators of compromise

### Stakeholder Communication



## **Internal Stakeholders**

- SOC Manager
- IT Team
- Legal Team
- Management

## **External Stakeholders**

- Customers
- Regulatory authorities
- Third-party vendors

## **3.3 SOAR and Automation**

Security Orchestration, Automation, and Response (SOAR) platforms automate repetitive security tasks.

## **Common SOAR Functions**

| <b>Function</b>  | <b>Example</b>              |
|------------------|-----------------------------|
| Ticket creation  | Auto-create incident ticket |
| Alert enrichment | Add IOC context             |
| Notification     | Send escalation email       |
| Containment      | Block malicious IP          |